台灣聯合大學系統107學年度學士班轉學生考試試題

科目 微積分　類組別 A3 A4 A7　共 1 頁 第 1 頁

甲、填充題：共 8 題，每題 8 分，共 64 分。請在答案卷上列出題號依序作答。

請注意：本（甲）部分，共 8 題，命題型態為填充題，必須以填充題形式將答案寫在答案卷第一頁，倘若答案被包含在演算過程中，將被視為試算草稿，無法採計計分。

參考用

1. Find the critical number of $y = 1 - \frac{4}{\pi^2}(\tan^{-1} x)^2$. Answer :________

2. Determine the limits of integration where $a \le b$ such that $\int_a^b (x^2 - 16)\, dx$ has minimal value. Answer :________

3. Evaluate $\int_{-\infty}^{\infty} \frac{e^x}{1+e^{2x}}\, dx$. Answer :________

4. Find the slope of the surface $f(x,y) = (x^3+y^3)^{1/3}$ at the point $(0,0)$ in the y-direction. Answer :________

5. Find the surface area of the portion of the plane $z = 4 - 2x - 2y$ that lies above the circle $x^2 + y^2 \le 1$ in the first quadrant. Answer :________

6. Find an equation of the tangent plane to the paraboloid $\mathbf{r}(u,v) = u\,\mathbf{i} + v\,\mathbf{j} + (u^2+v^2)\,\mathbf{k}$ at the point $(1,2,5)$. Answer :________

7. Evaluate the integral $\int_0^{\infty}\int_0^{\infty} \frac{1}{(1+x^2+y^2)^2}\, dxdy$. Answer :________

8. Use a change of variables to find the volume of the solid region lying below the surface $z = \sqrt{(x+4y)(x-y)}$ and above the plane region R : region bounded by the parallelogram with vertices $(0,0), (1,1), (5,0)$ and $(4,-1)$. Answer :________

乙、計算、證明題：共 3 大題，每大題 12 分，共 36 分。須詳細寫出計算及證明過程，否則不予計分。

1. A heat-seeking particle is located at the point $(-1,2)$ on a metal plate whose temperature at (x,y) is $T(x,y) = 64 - 2x^2 - y^2$. (a) (6 分) In what direction from $(-1,2)$ does the temperature increase most rapidly? What is this rate of increase? (b) (6 分) Find the path of the particle as it continuously moves in the direction of maximum temperature increase.

2. Determine if the given series converges or diverges. Explain your reasoning.

a. (6 分) $\sum_{n=1}^{\infty} \left(\frac{3n+2}{n+3}\right)^n$　　b. (6 分) $\sum_{n=1}^{\infty} \frac{e^{2/n}}{n^2}$

3. Find the maximum value of $\int_C y^3\, dx + (27x - x^3)\, dy$, where C is any circle in the xy-plane, oriented counterclockwise.